# BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
# PRACTICAL JOURNAL

**Class: TYIT** **Sem: V** **Roll No.: TYIT58** **Date: 27/01/2026**

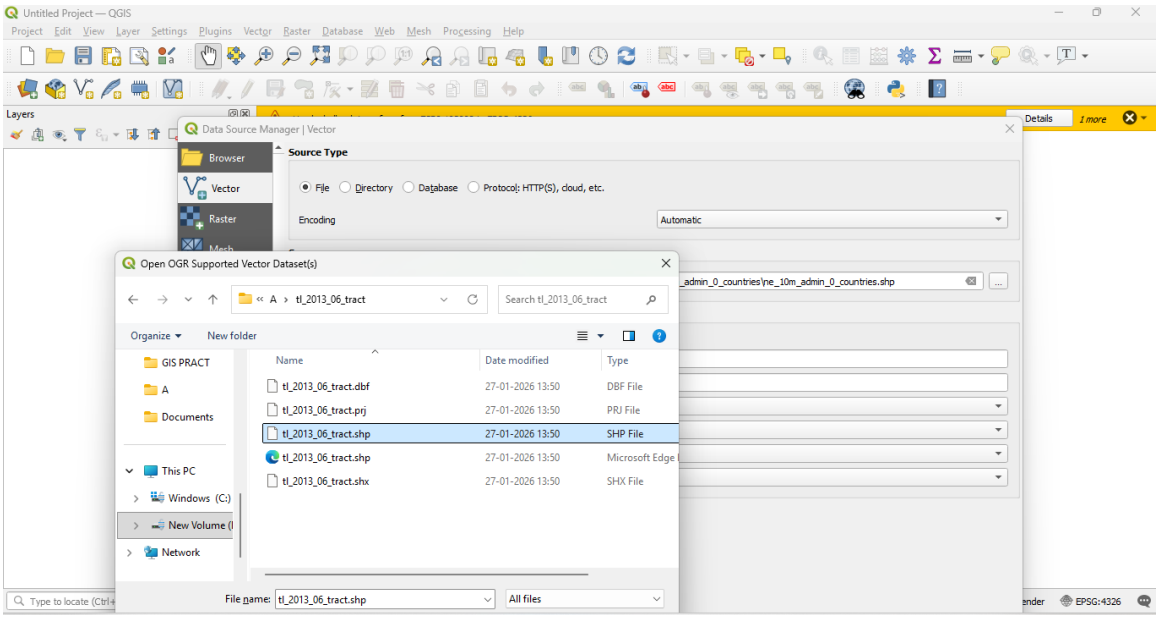
**Course Name: Geographic Information System (GIS)** **Page no:**

**Practical Number :- 07**

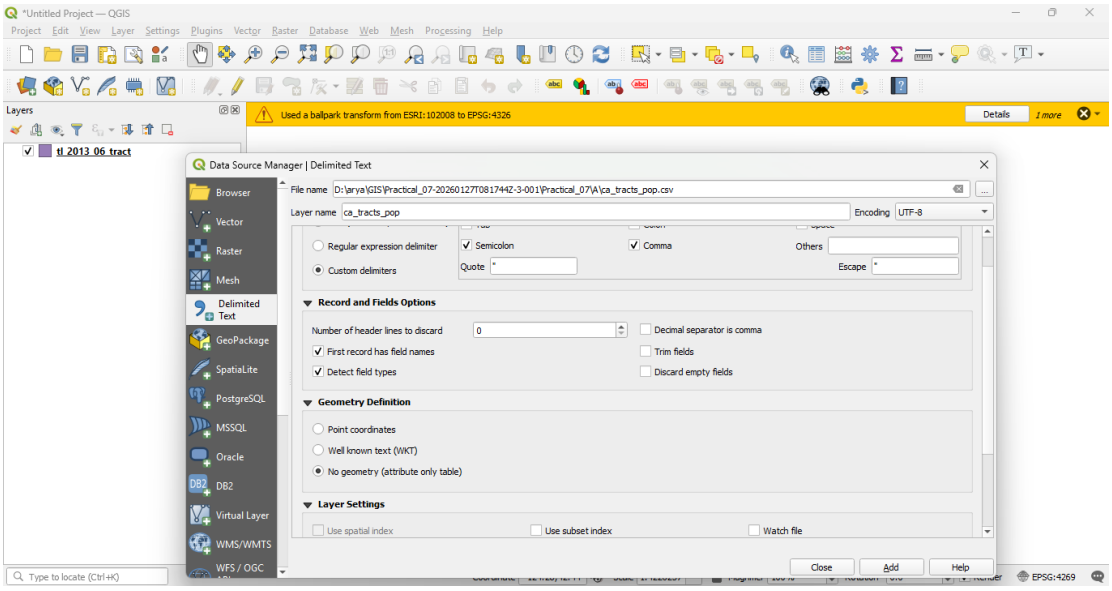
## Q] Managing Data Tables and Spatial Data Sets:

### A.Table joins

- Start a new project
- Go to Layer → Add Layer → Add new Vector Layer "I:\GIS_Workshop\Practicals\Practical_07\A\Data\tl_2013_06_tract.zip"




- Go to Layer → Add Layer → Add Delimited Text Layer And add I:\GIS_Workshop\Practicals\Practical_07\A\Data\ca_tacts_pop.csv"




- In the layer panel, Right click on "tl_2013_06_tract", layer and select Properties

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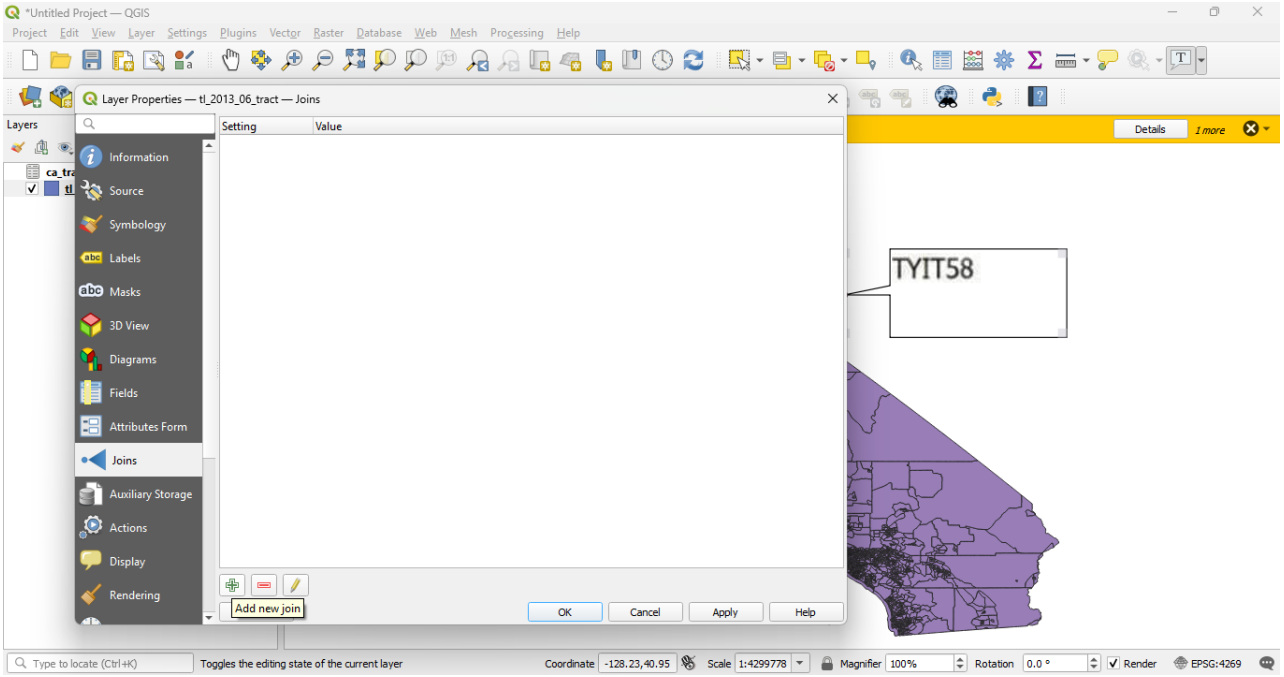
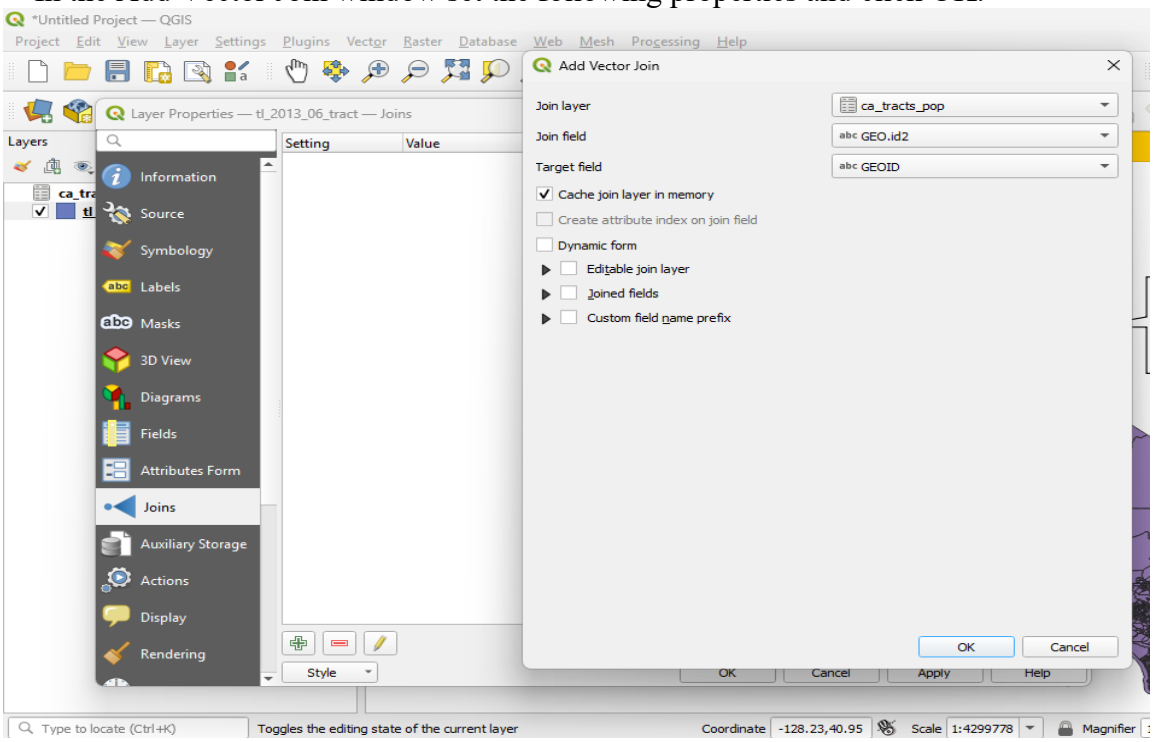



- Select the JOIN option in Properties, and click on "+" button to add new table join.
- In the Add Vector Join window set the following properties and click OK.

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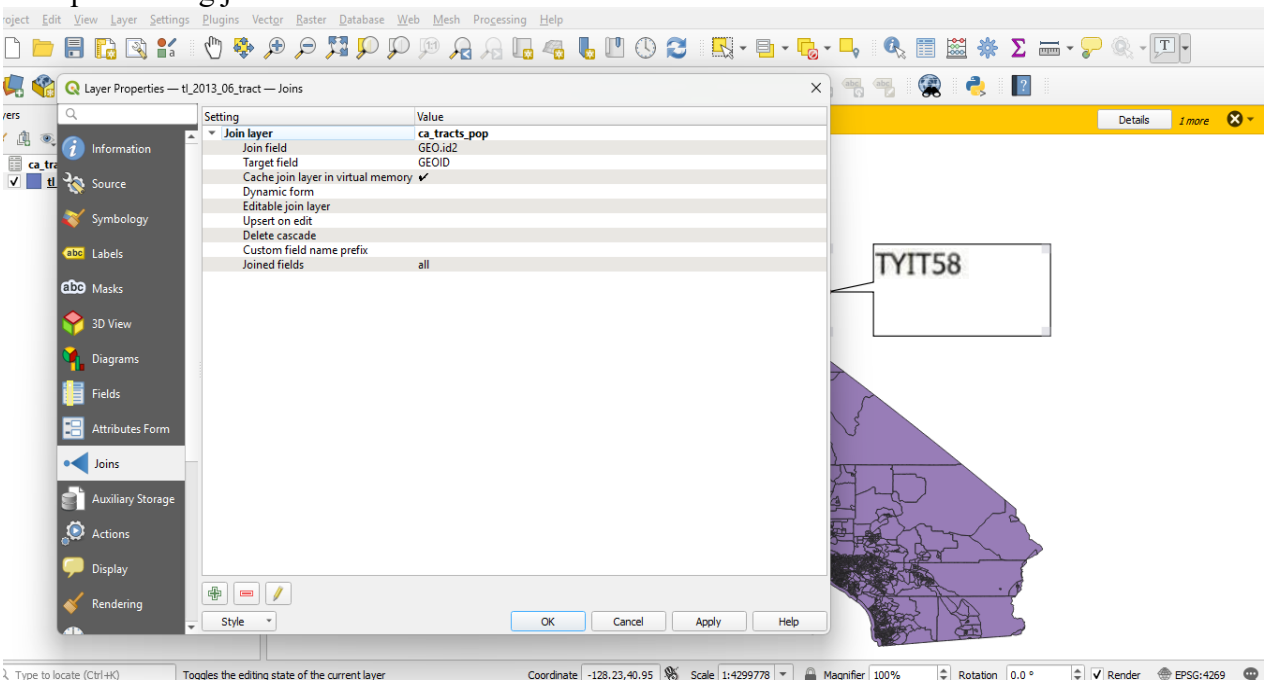
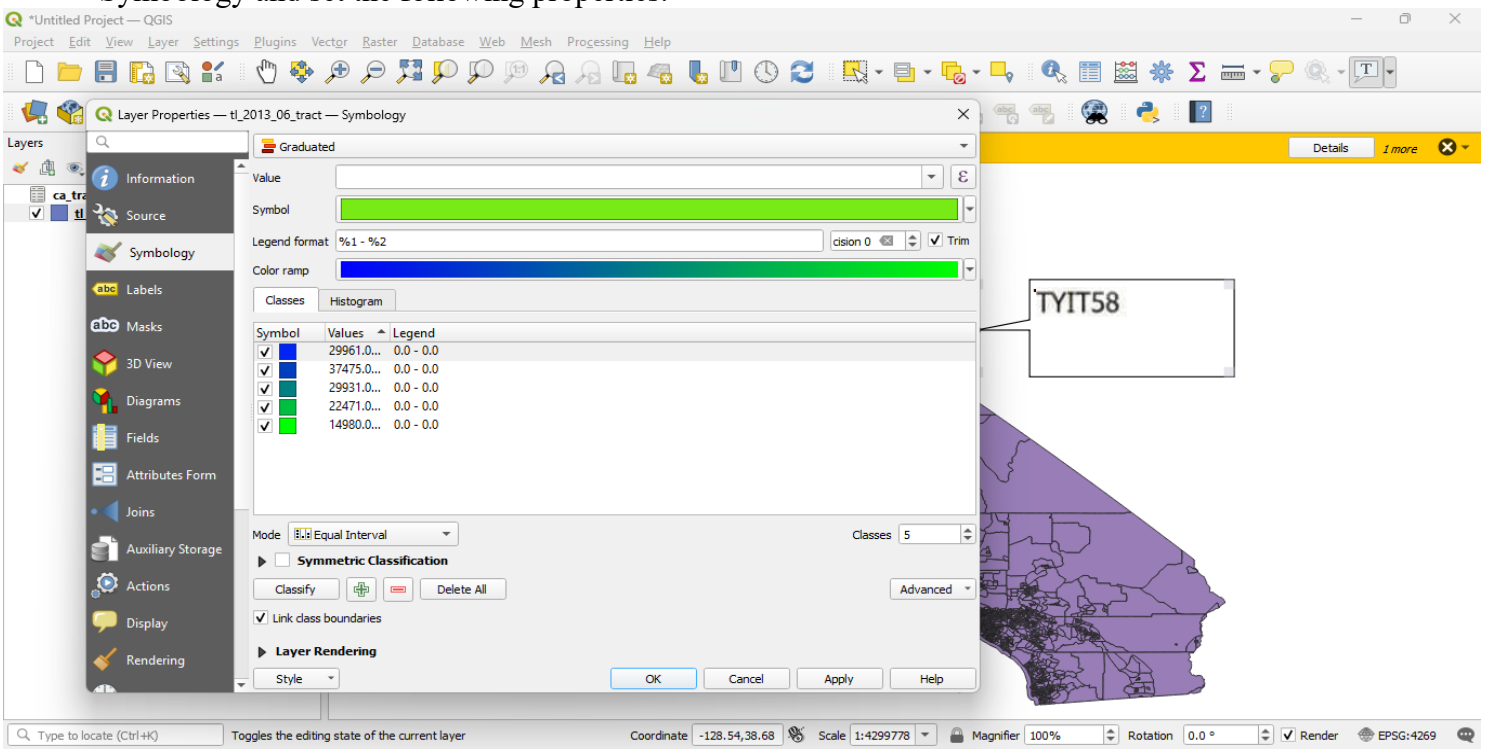
- After performing join




- For more clear output, select "tl_2013_06_tact" from Layer Panel, right click and select properties. Go to Symbology and set the following properties.

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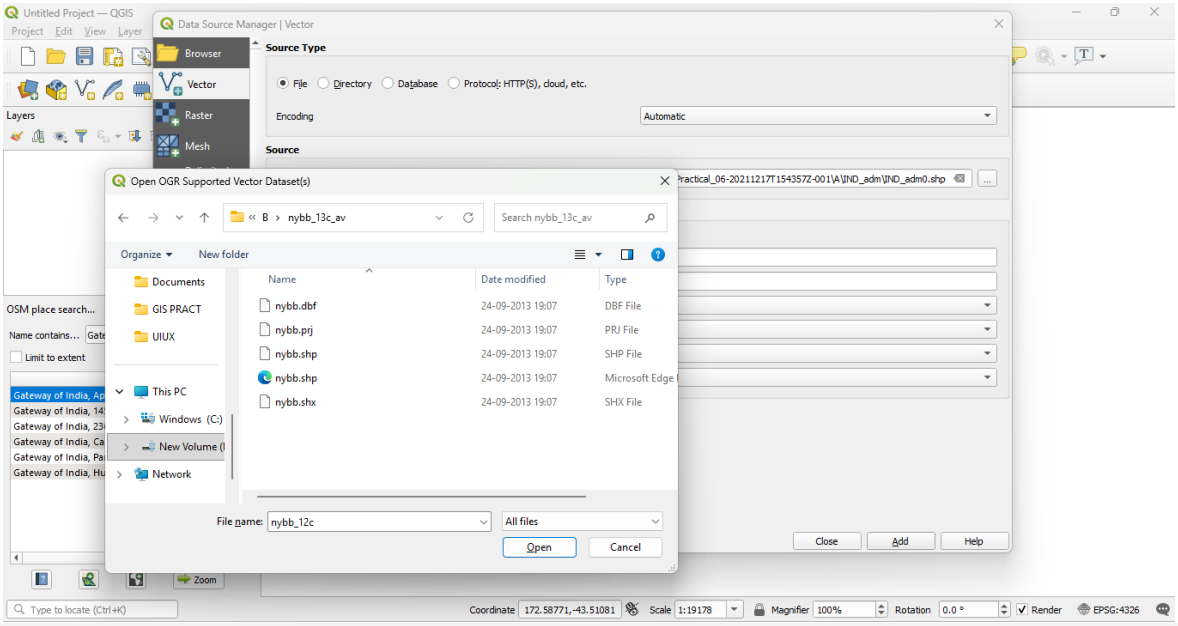
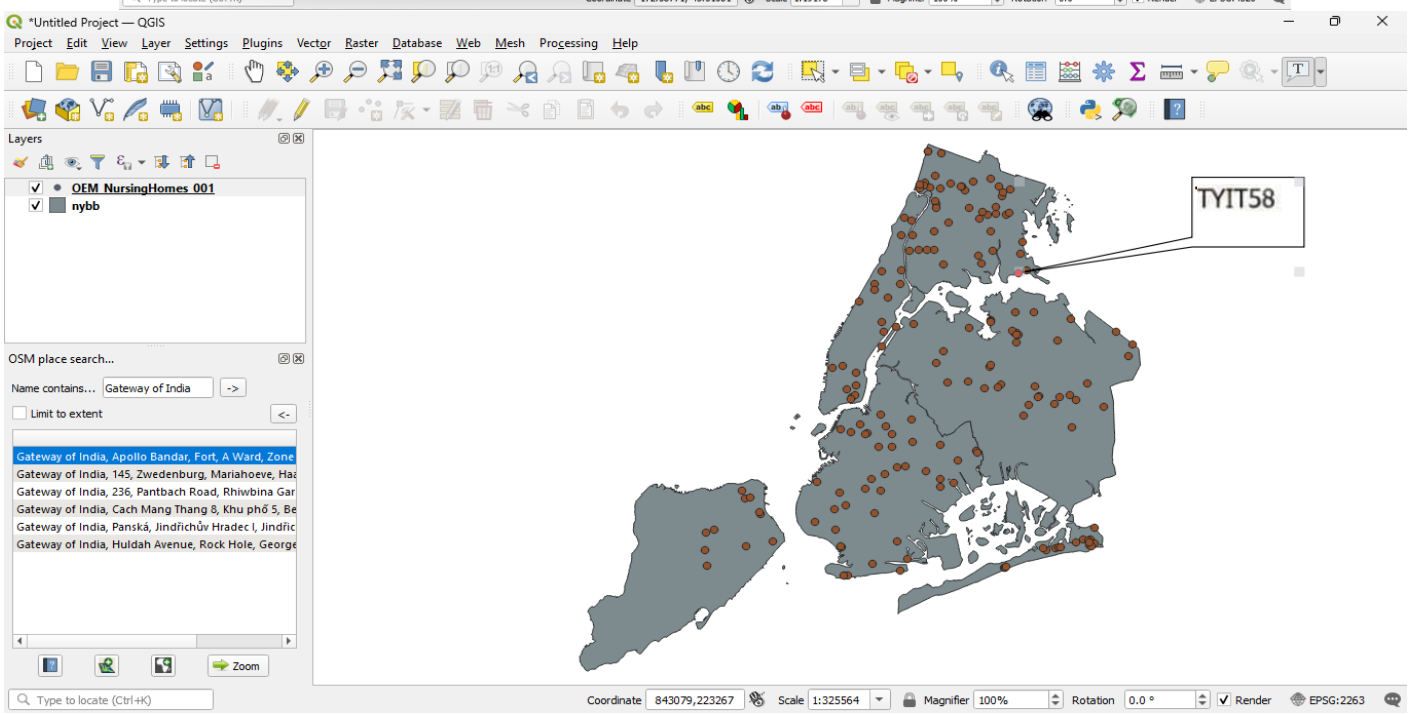
## B. Spatial Joins:

- Go to Layer → Add Layer → Add Vector Layer → Select "I:\GIS_Workshop\Practicals\Practical_07\B\Data\nybb_12c\nybb_13c_av\nybb.shp" and "I:\GIS_Workshop\Practicals\Practical_07\B\Data\OEM_NursingHomes_001\OEM_NursingHomes_001.shp", from data folder.




- Go to Vector → Data Management Tools → Join Attributes by Location

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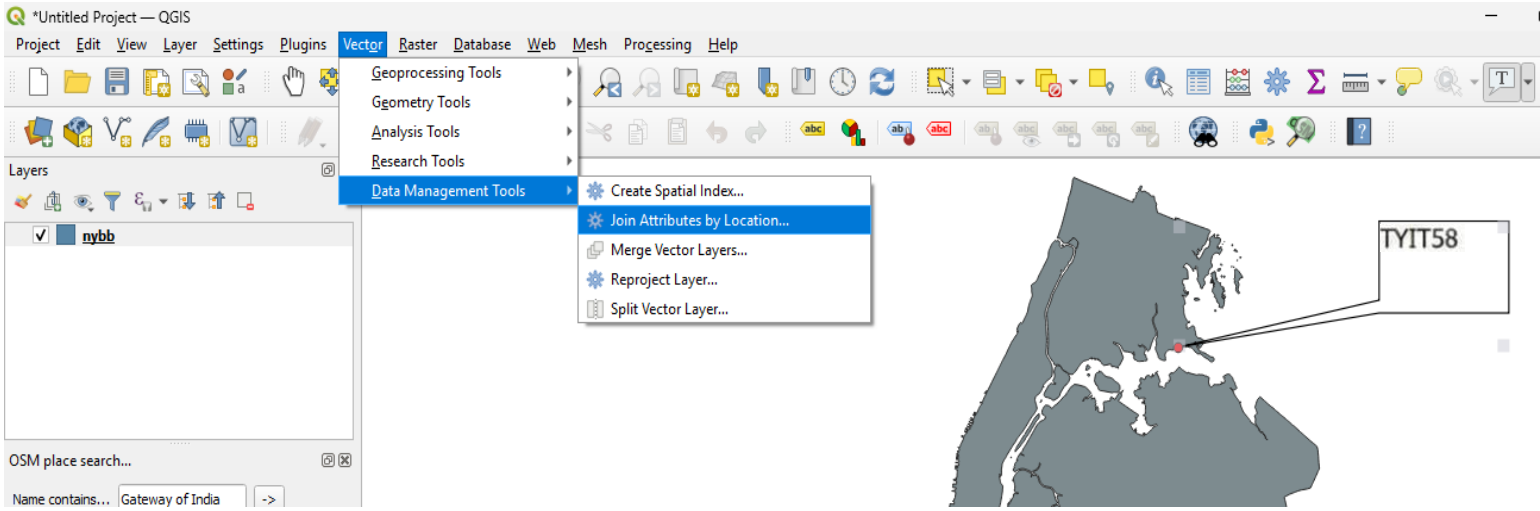


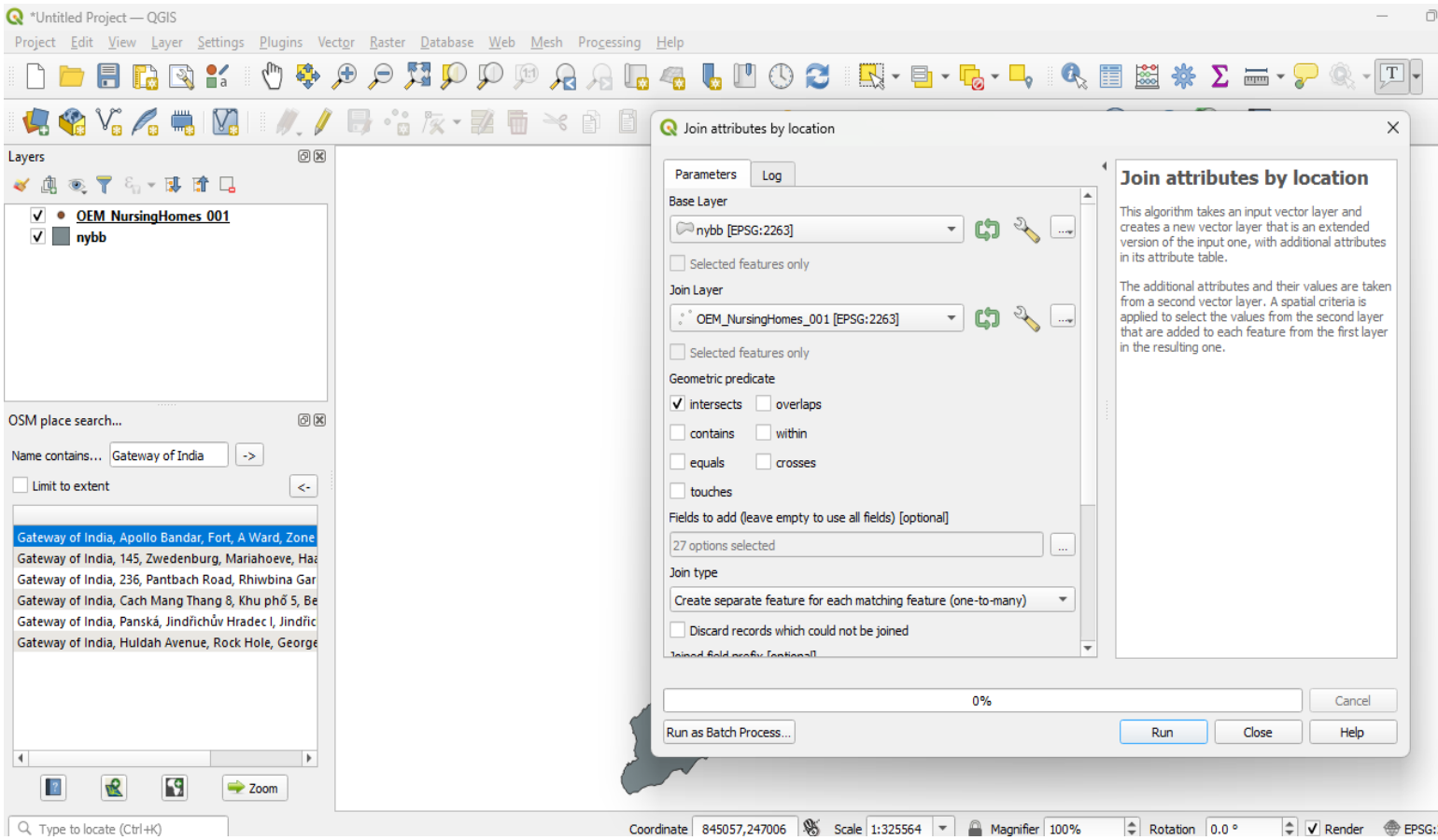


- Click on the three dots: Fields to select: select all

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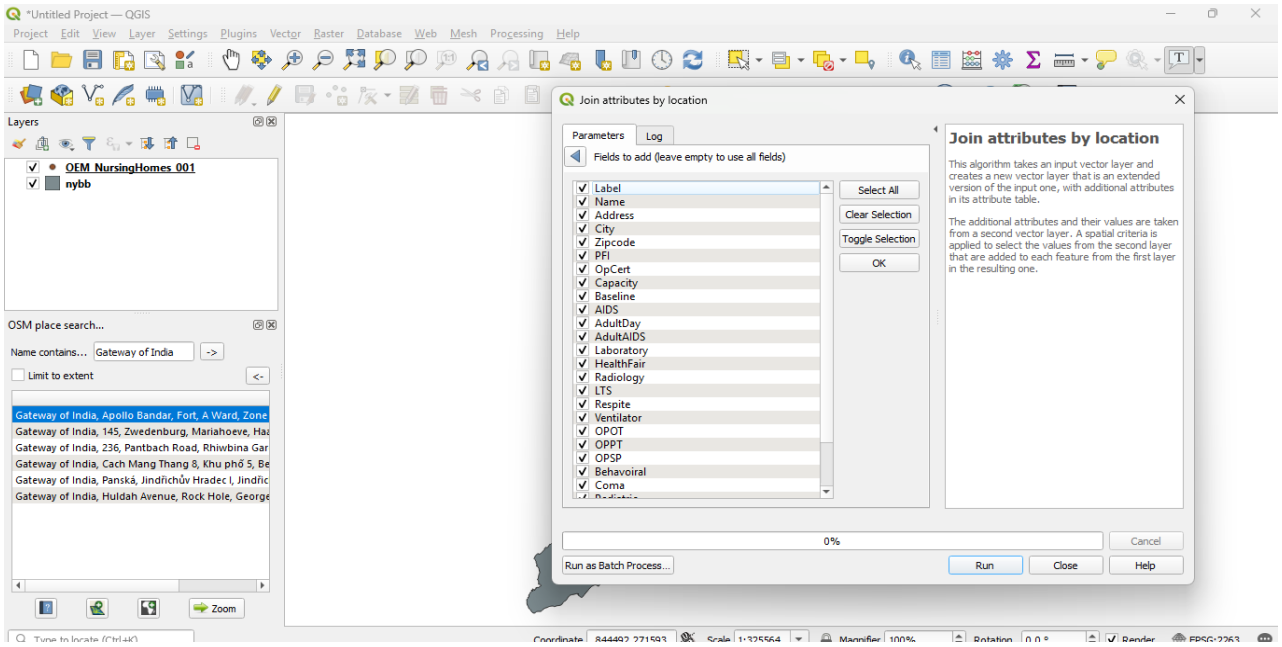
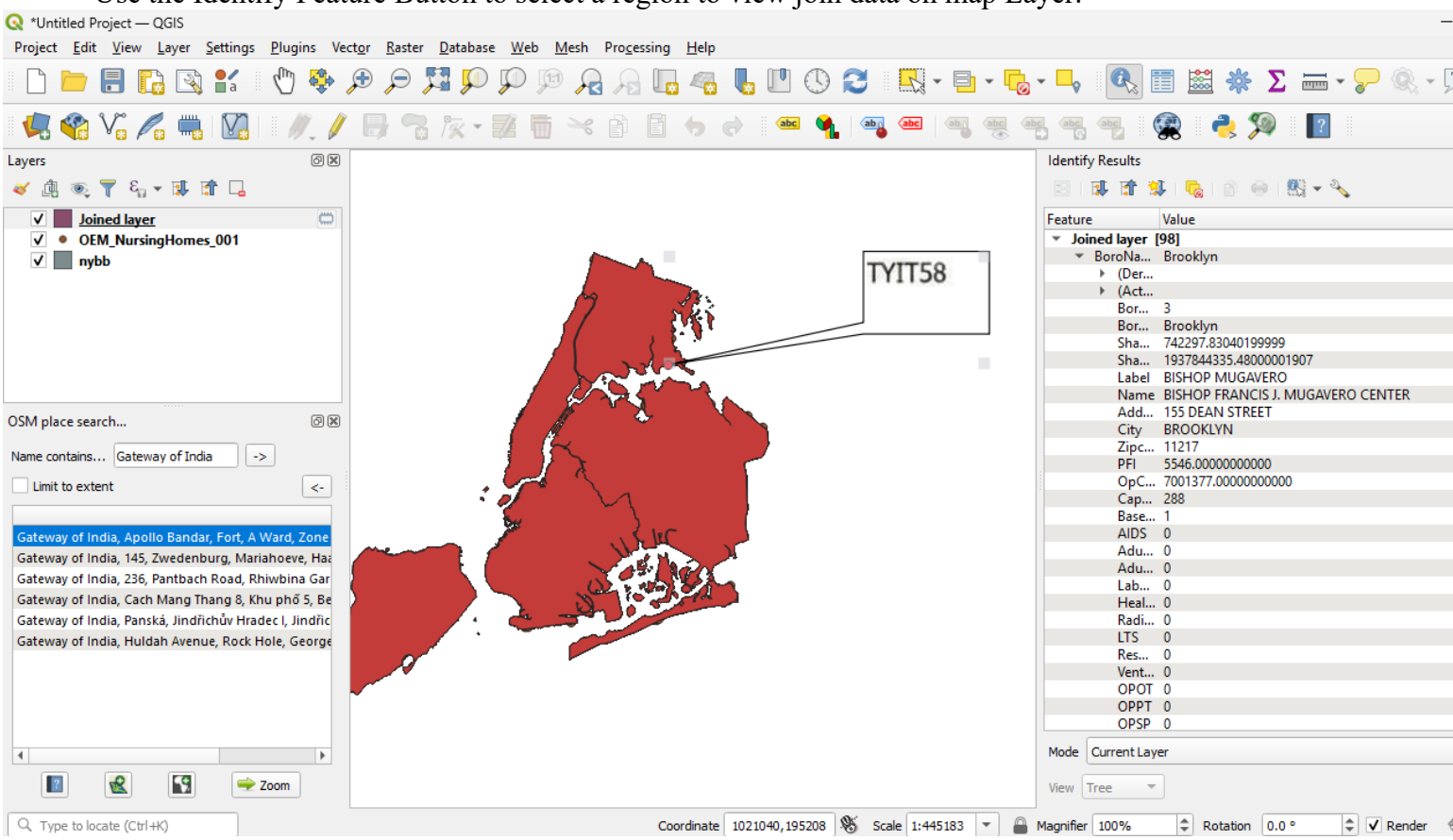



- Use the Identify Feature Button to select a region to view join data on map Layer.

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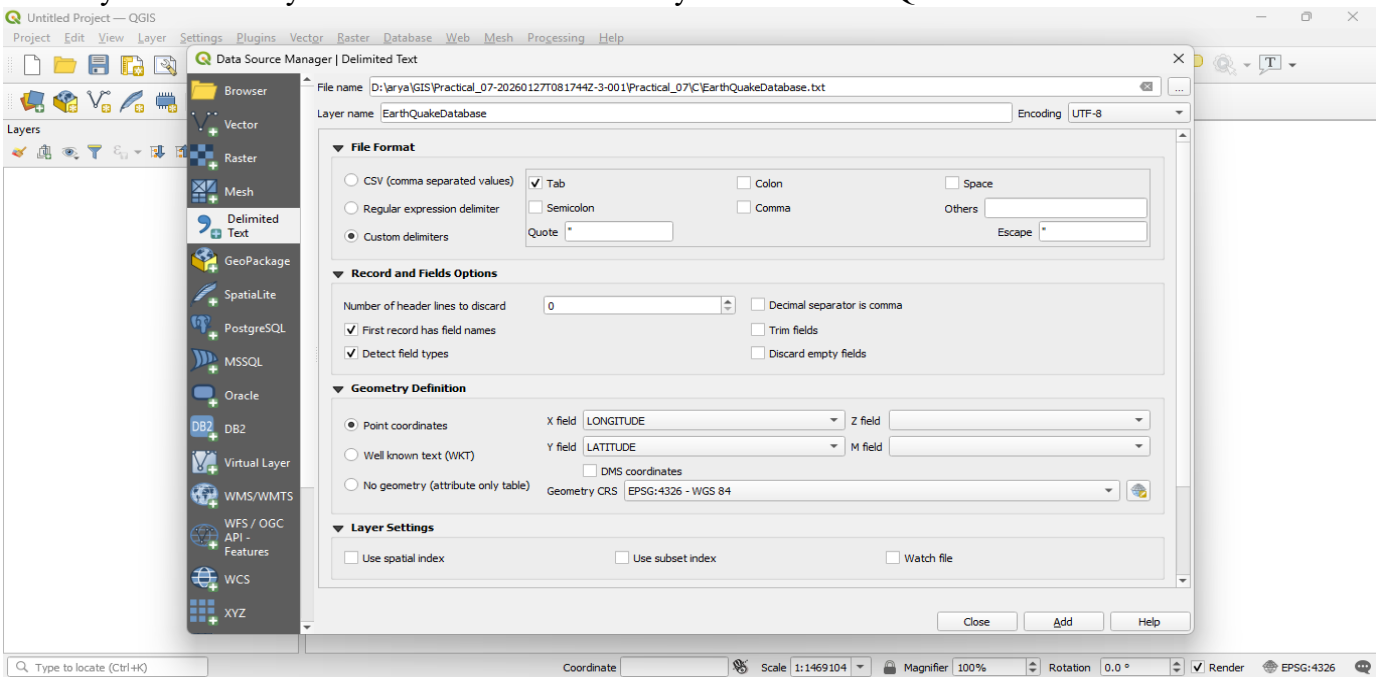
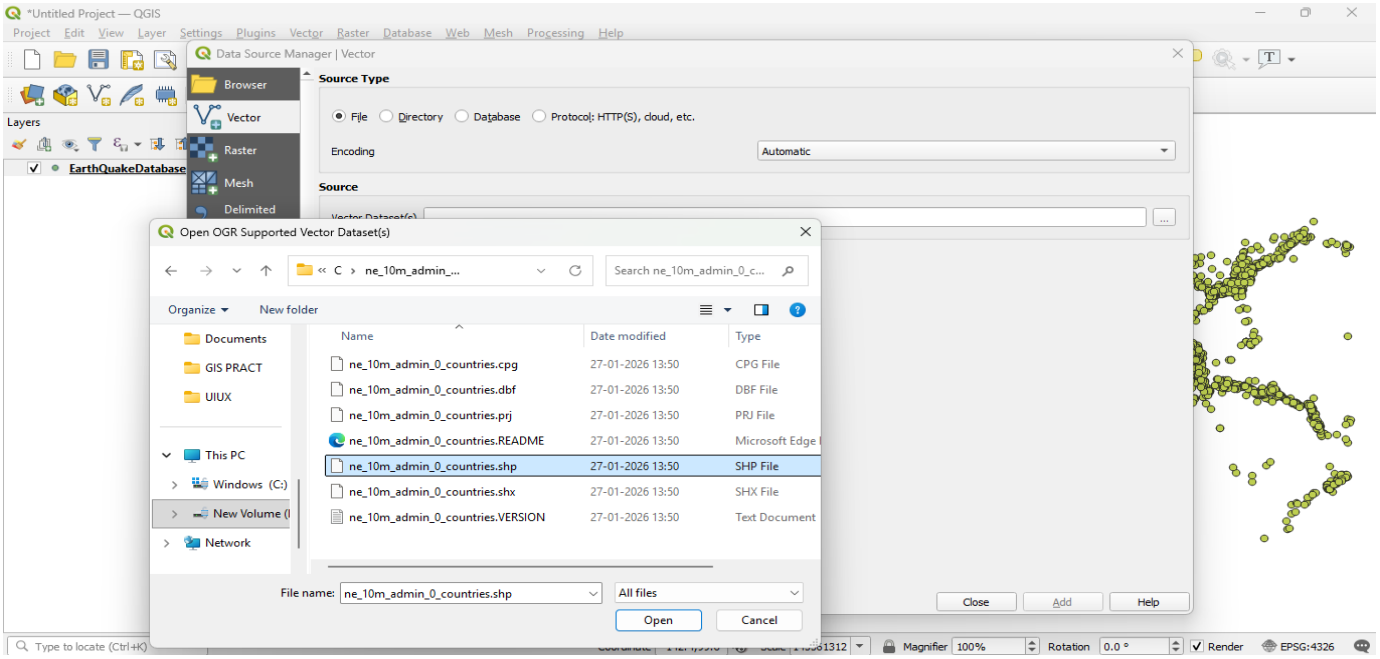
## C. Points in polygon analysis

- Points in polygon analysis
- Go to Layer → Add Layer → Add Delimited Text Layer Select "EarthQuakeDatabase.txt.




- Go to Layer → Add Layer → Add Delimited Text Layer "I:\GIS_Workshop\Practicals\Practical_07\C\Data\ne_10m_admin_0_countries.zip"

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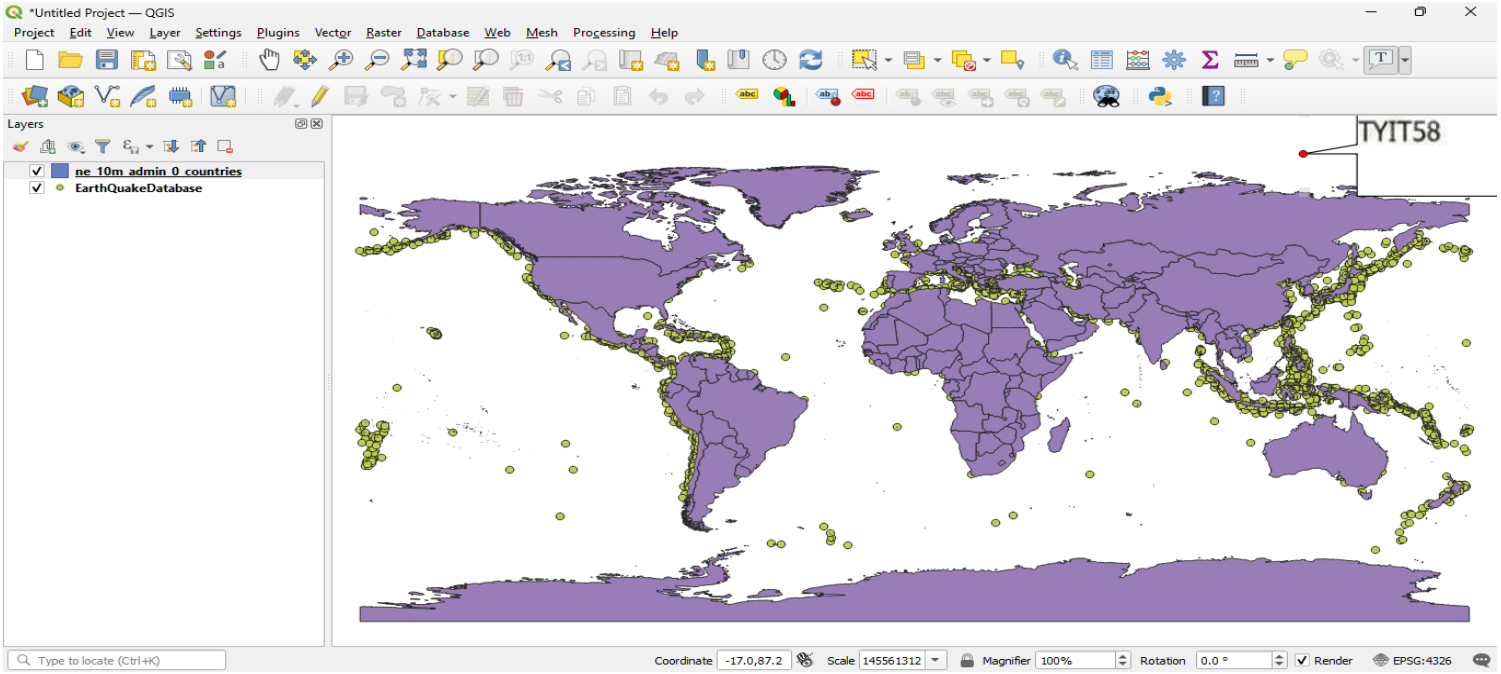
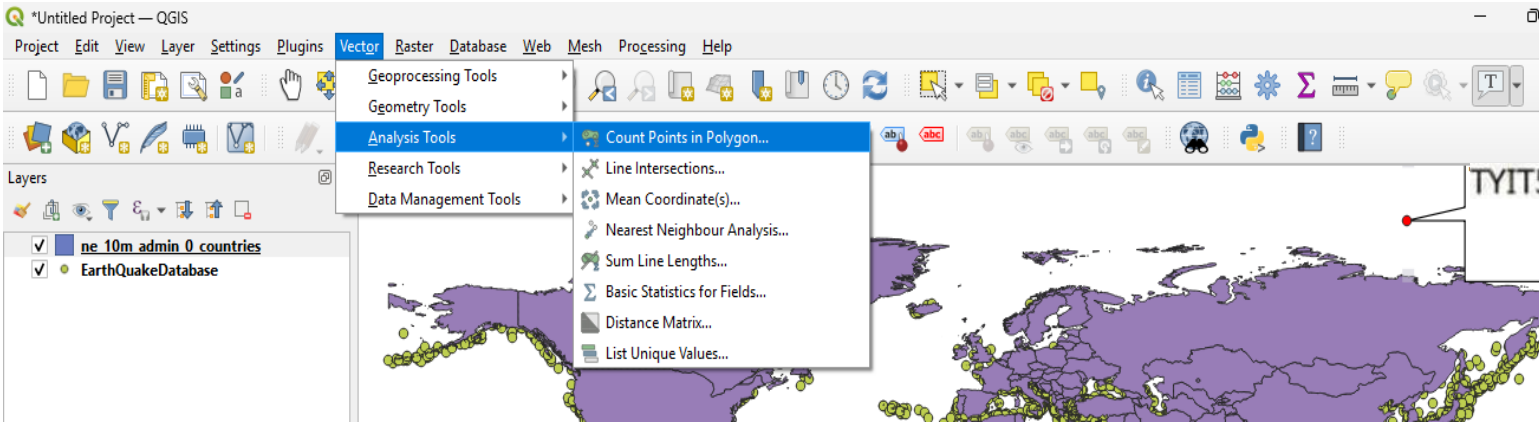



- Go to Analysis Tools -> Count Points in Polygon




- Make the Following Changes in the tab

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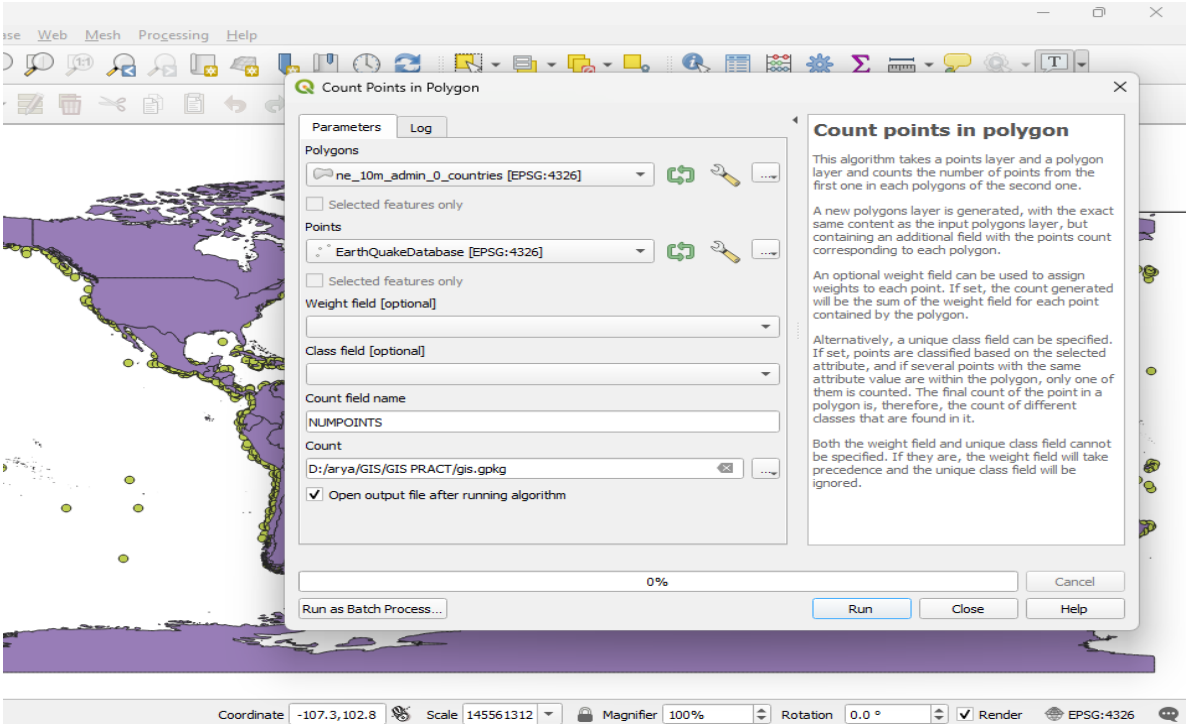
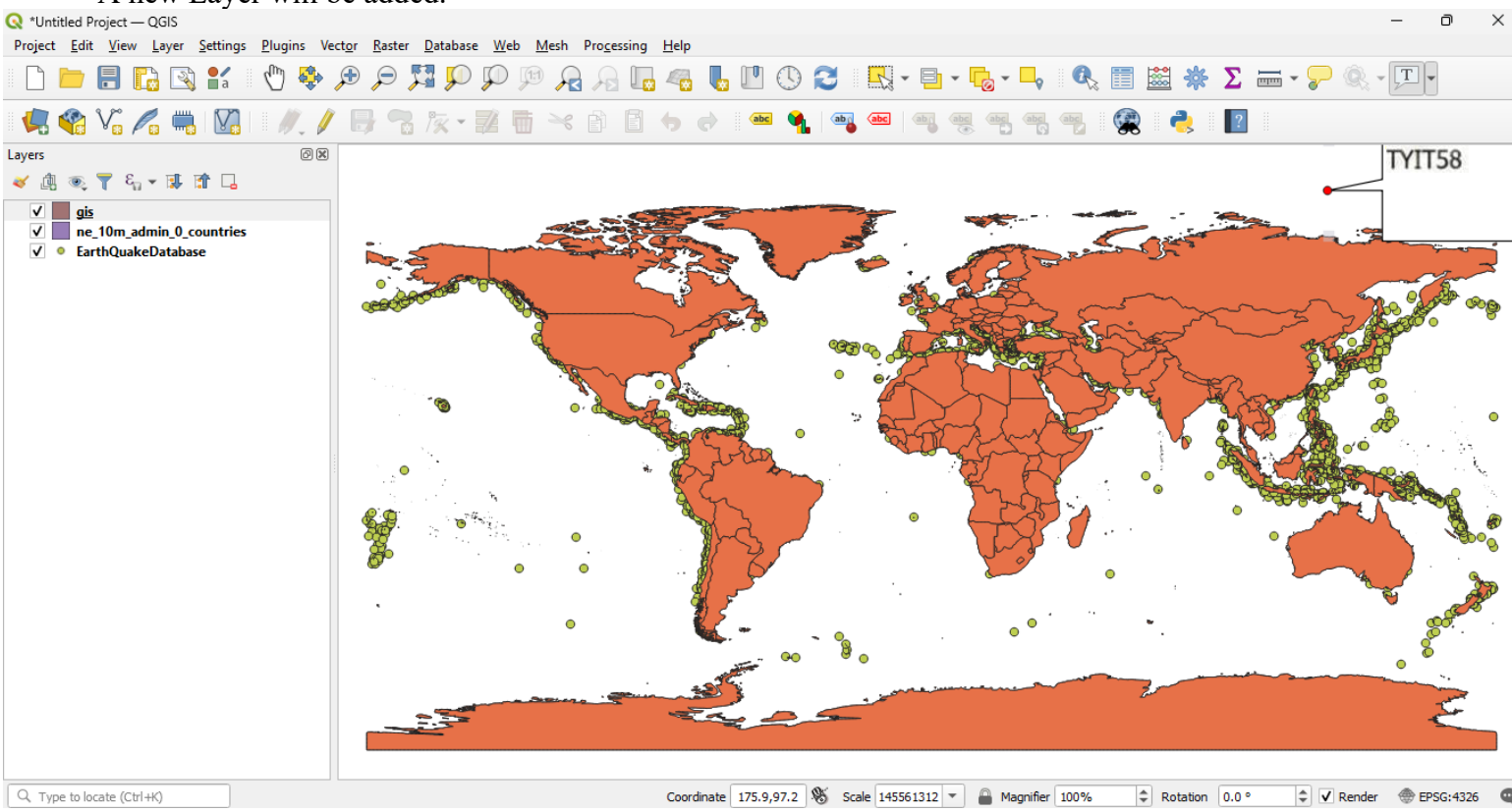



- A new Layer will be added.

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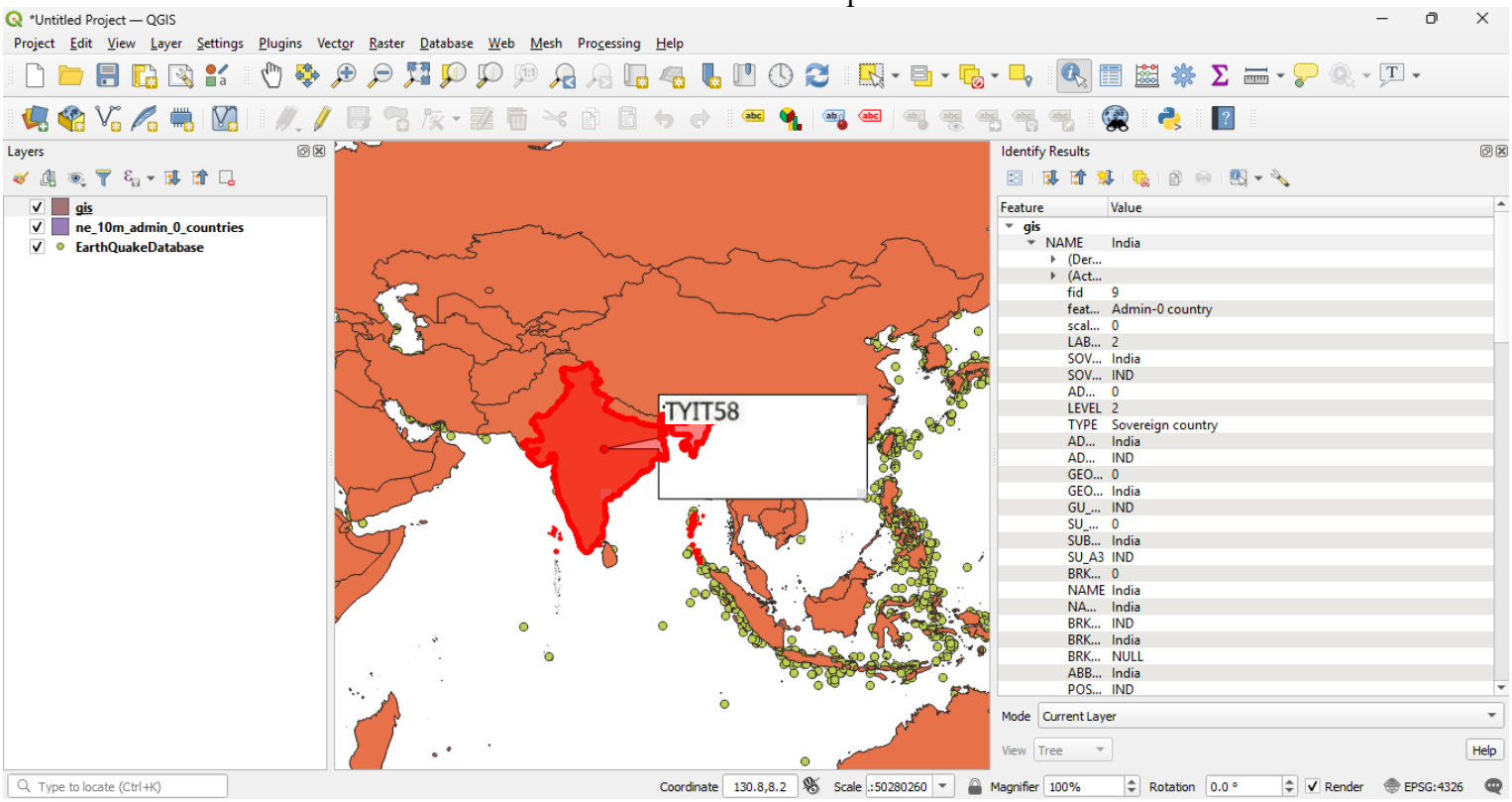
- Go to the information buttons -> Click on a location on map

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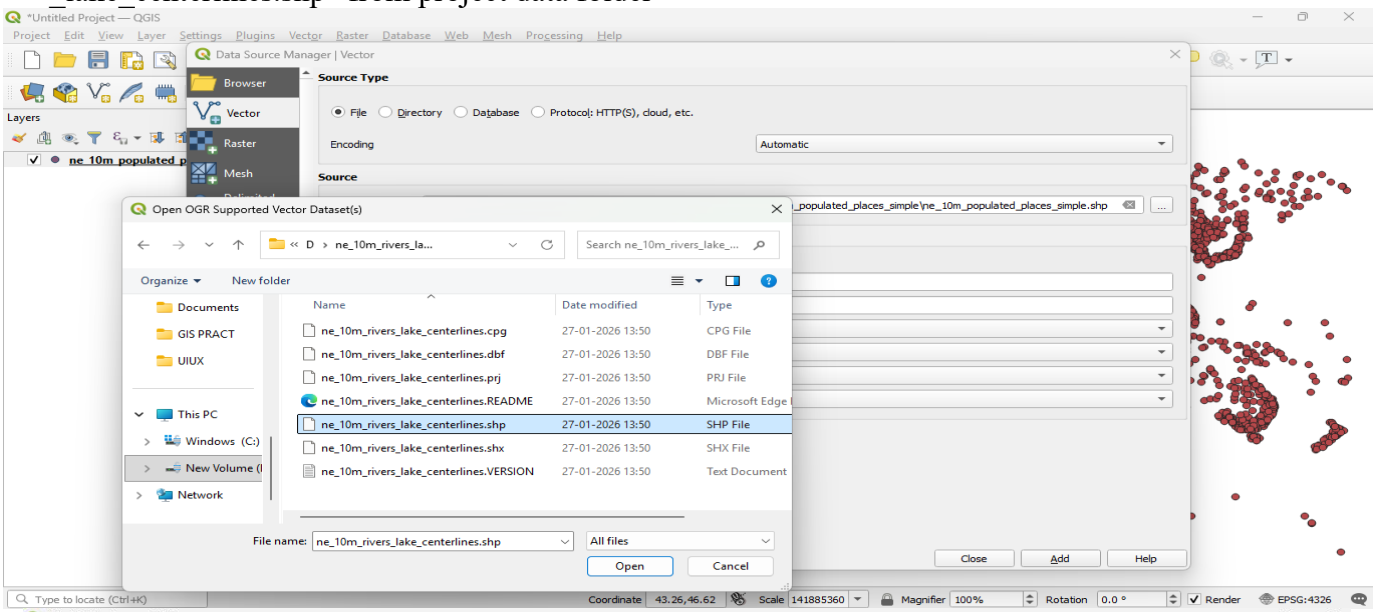
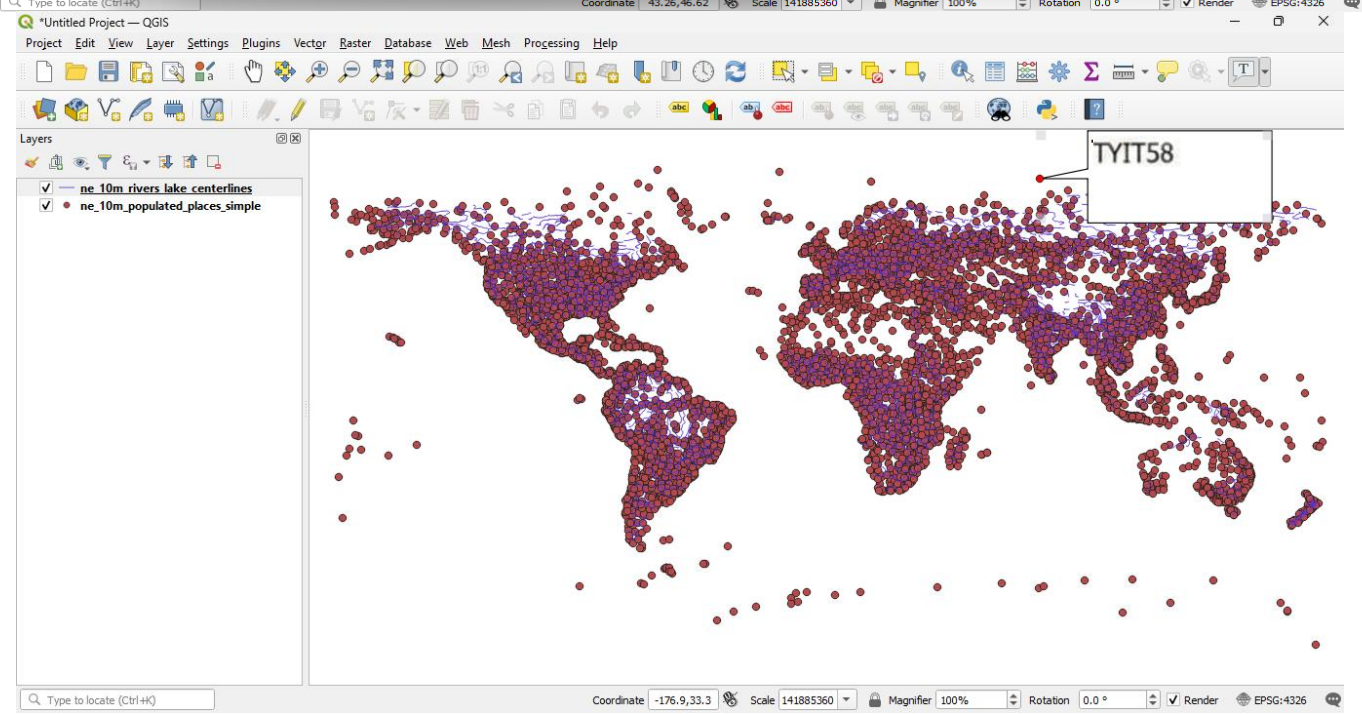
## D. Performing spatial queries

- Use the select Feature button to check country wise counting of Earthquakes.
- Performing spatial queries
- Go to Layer → Add Layer → Add Vector Layer and load "\GIS_Workshop\Practicals\Practical_07\D\Data\ne_10m_populated_places_simple\ne_10m_populated_places_simple.shp" and "I:\GIS_Workshop\Practicals\Practical_07\D\Data\ne_10m_rivers_lake_centerlines\ne_10m_rivers_lake_centerlines.shp" from project data folder

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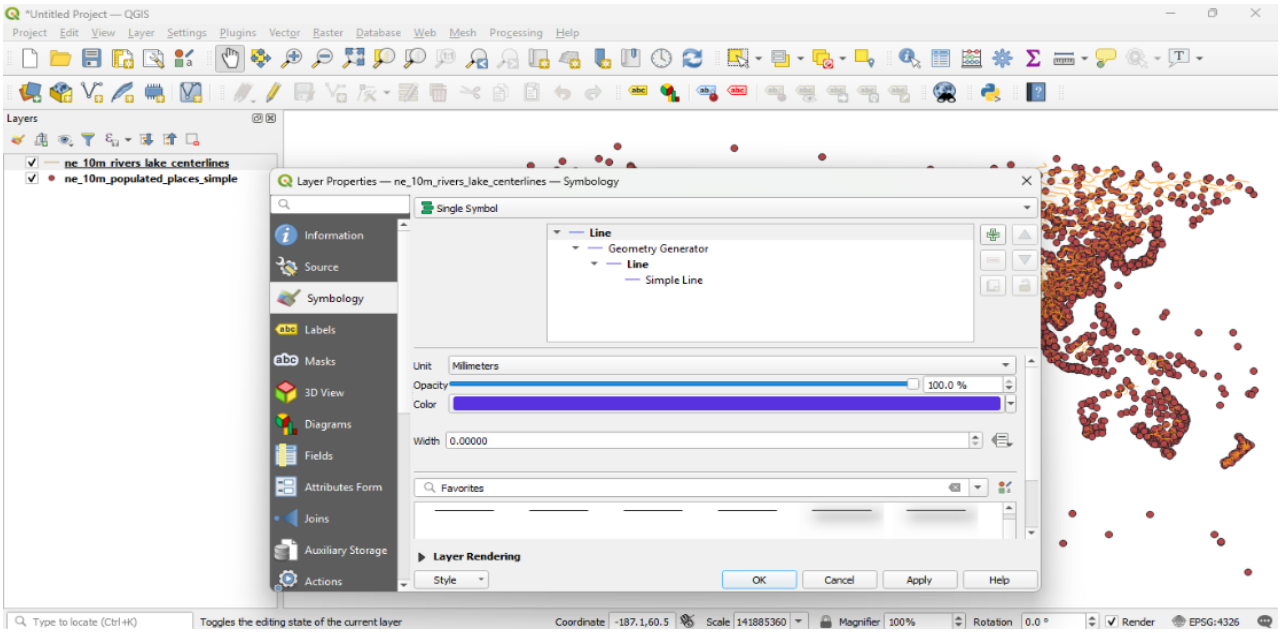
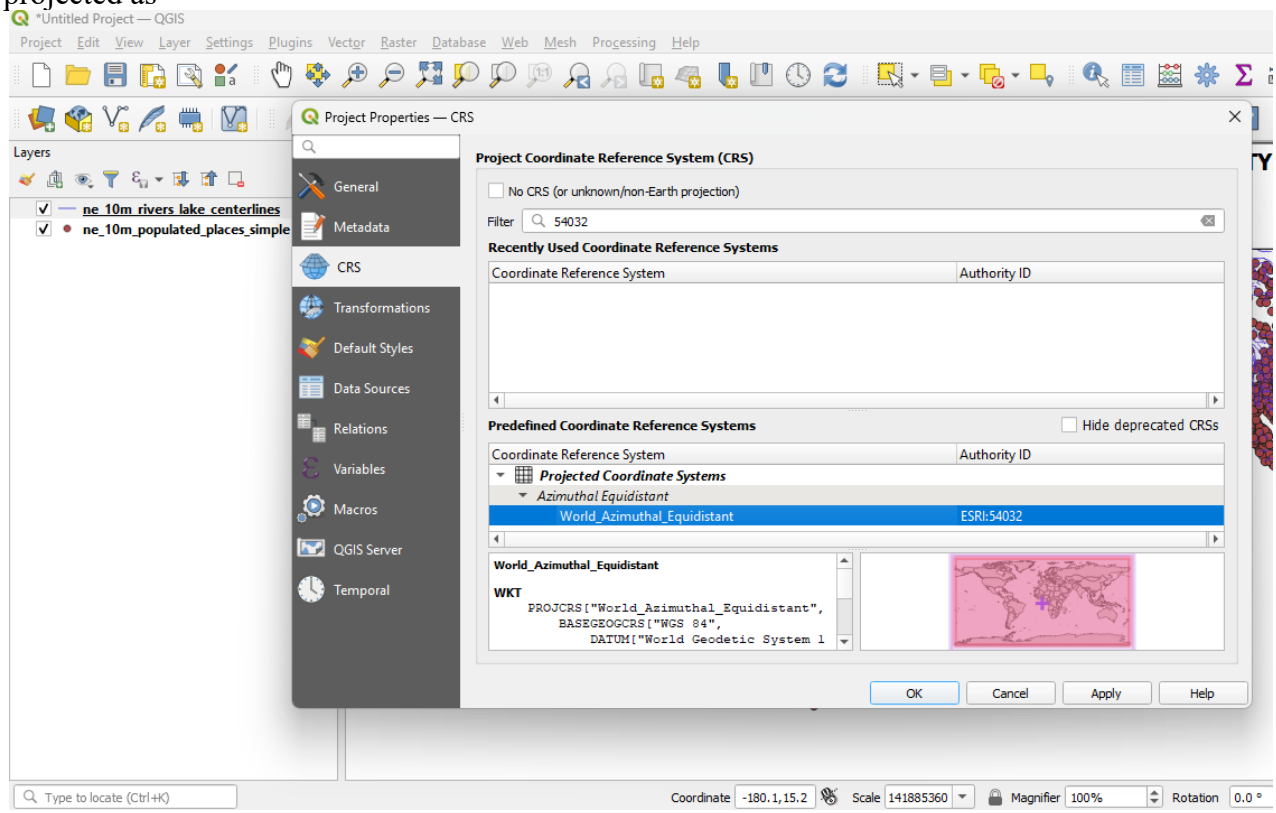
- Go to Properties of River_lakes centerlines and change the symbol.




- Open project Properties → Set CRS "World_Azimuthal_Equidistant EPSG 54032" . The map will be re-projected as

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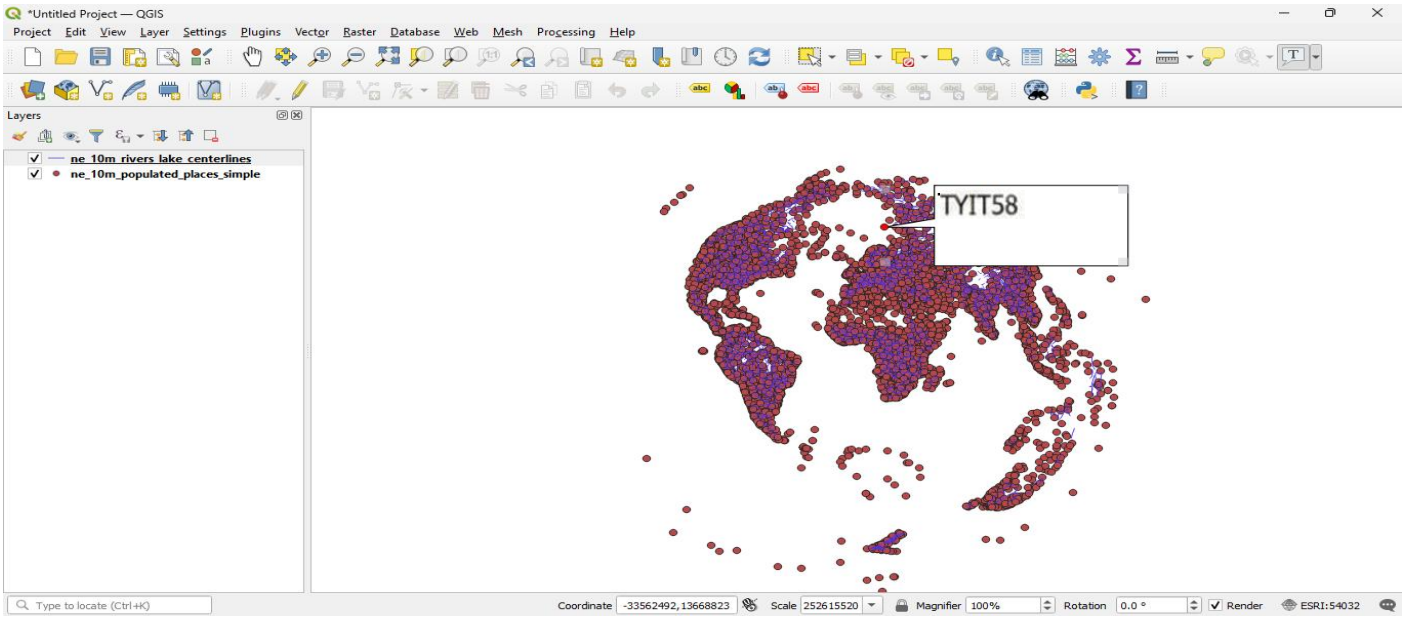
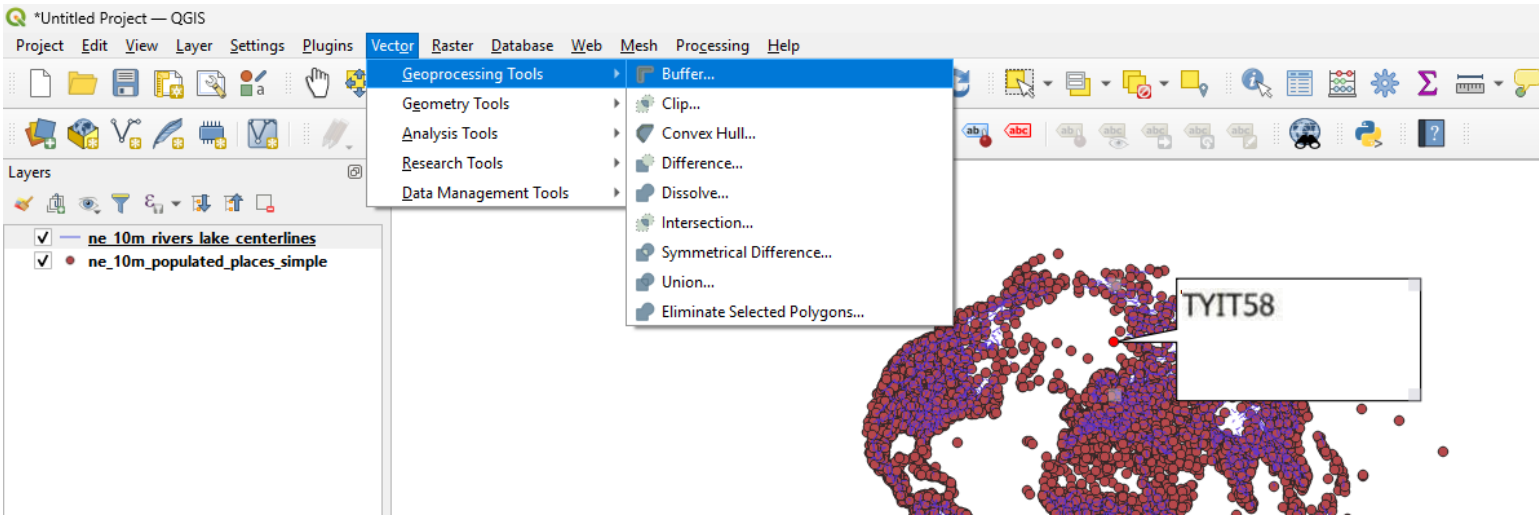



- Go to Vector → Geoprocessing Tool → Buffer

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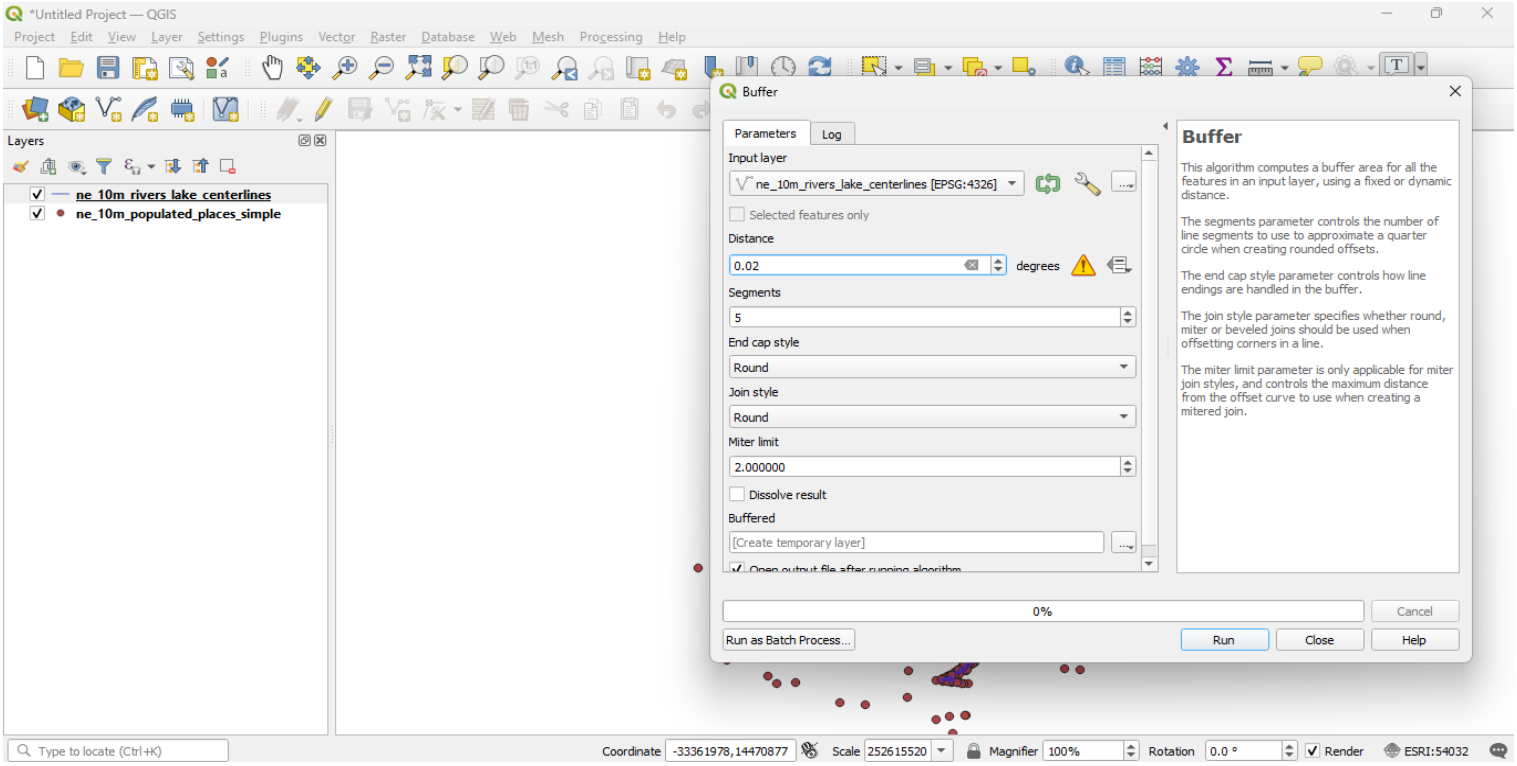
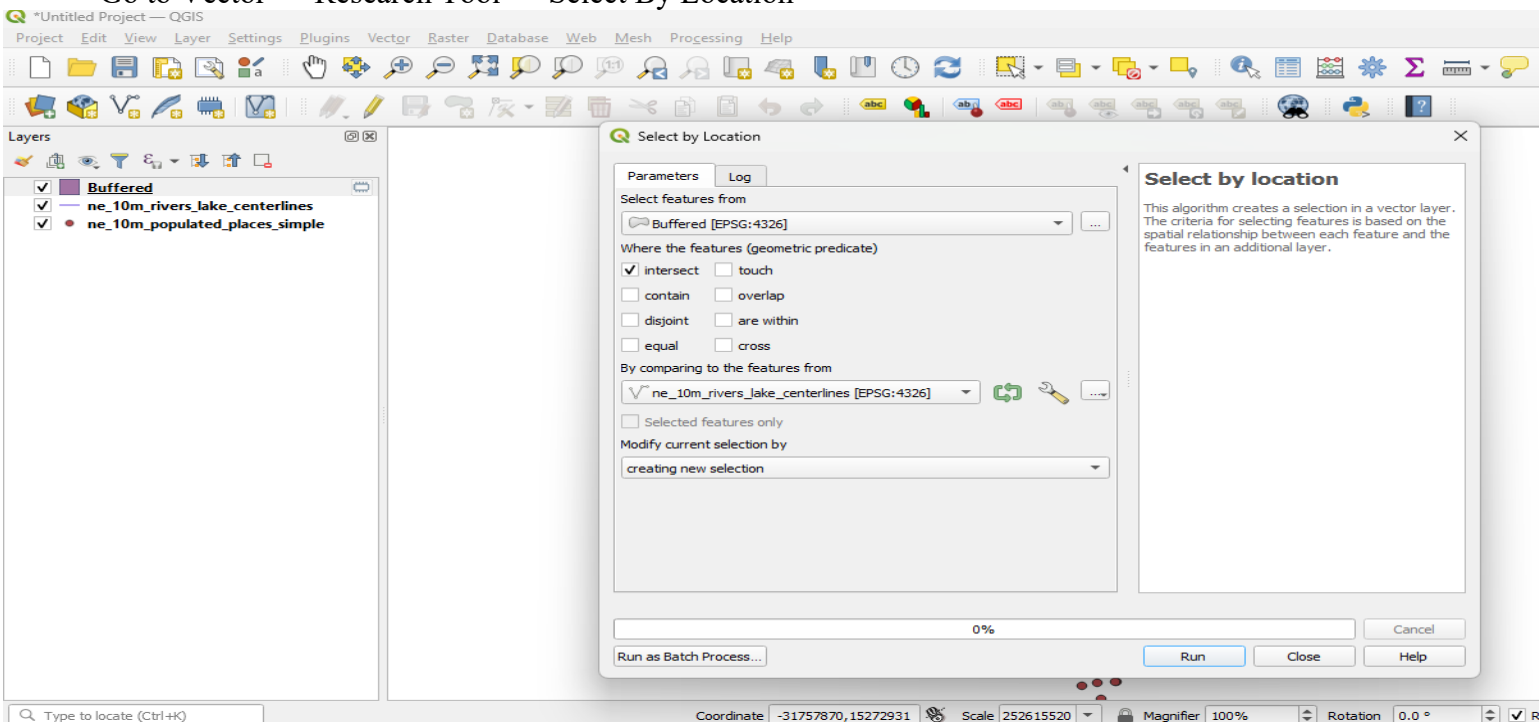



- Create a buffer for River
- Go to Vector → Research Tool → Select By Location

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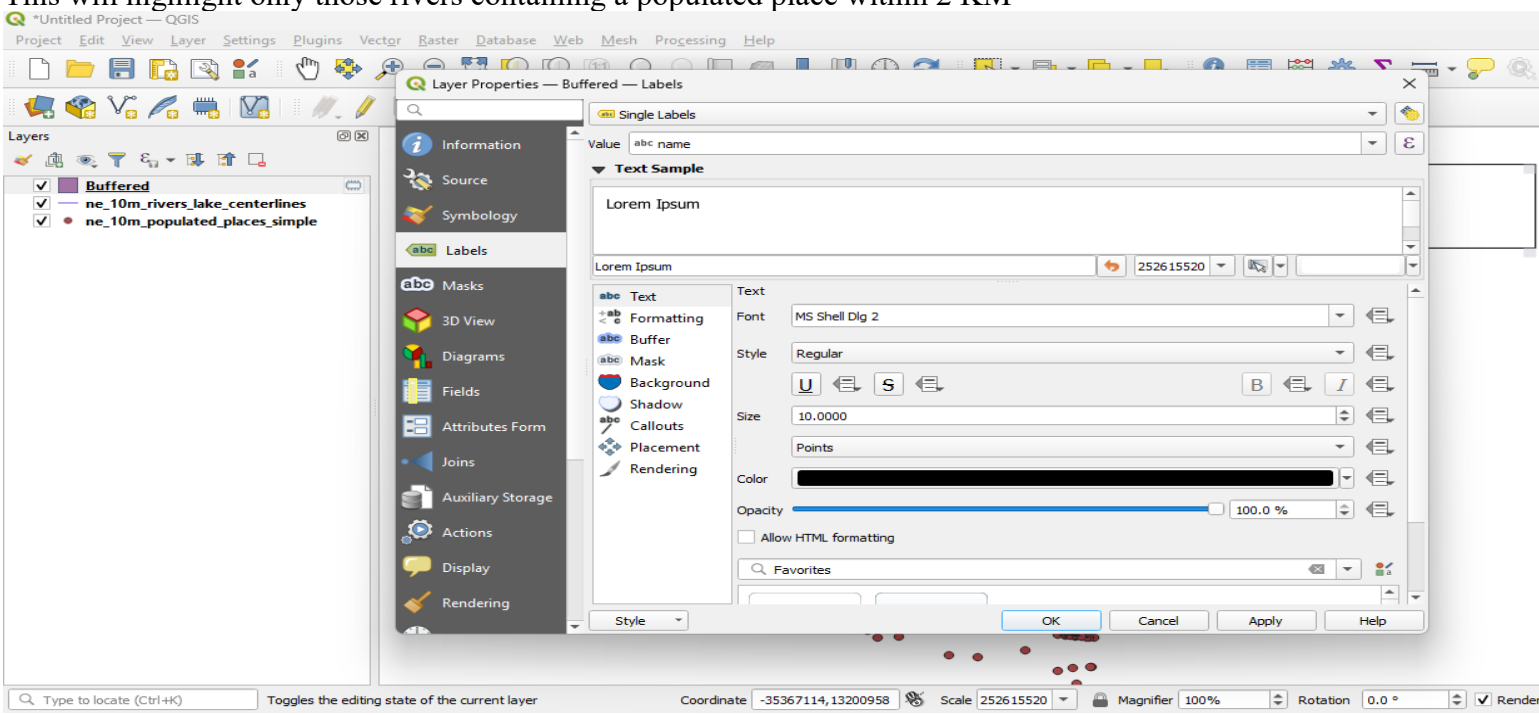
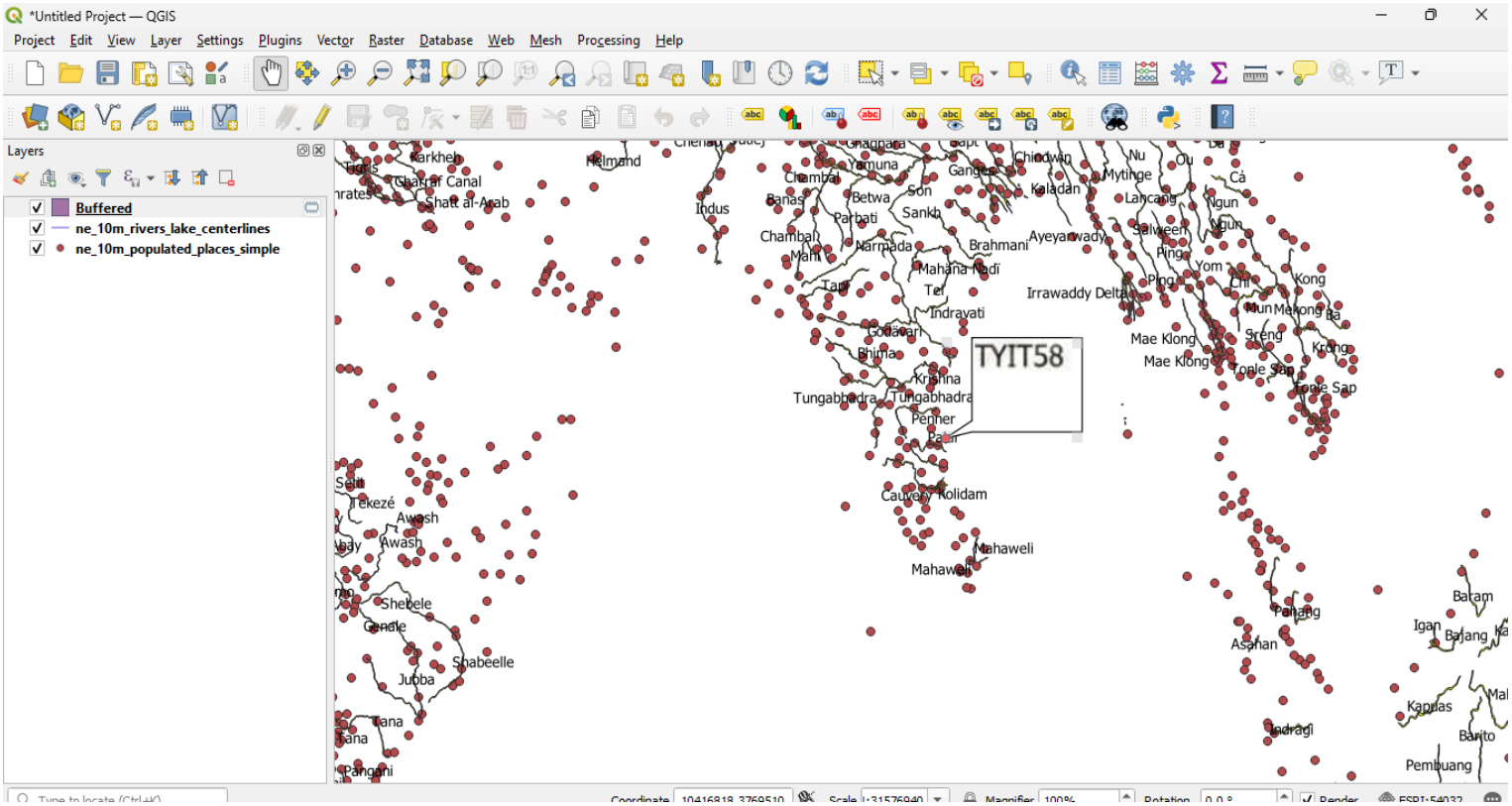
This will highlight only those rivers containing a populated place within 2 KM

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